SILATIO 20 mg Tablet

Each film coated tablet contains:
Sildenafil citrate eq. to
Sildenafil 20 mg

PRODUCT DESCRIPTION:

White, round, biconvex, film coated tablet with engraved 20 on one side and scored on the other side
The tablet is not meant to be split and should be swallowed whole.

MECHANISM OF ACTION:

Pharmacology

Sildenafil is a potent and selective inhibitor of cyclic guanosine monophosphate (cGMP) specific phosphodiesterase type 5 (PDE5), the enzyme that is responsible for degradation of cGMP. Apart from the presence of this enzyme in the corpus cavernosum of the penis, PDE5 is also present in the pulmonary vasculature. Sildenafil, therefore, increases cGMP within pulmonary vascular smooth muscle cells resulting in relaxation. In patients with pulmonary arterial hypertension this can lead to vasodilation of the pulmonary vascular bed and, to a lesser degree, vasodilatation in the systemic circulation.

Pharmacokinetics

Absorption

Sildenafil is rapidly absorbed. Maximum observed plasma concentrations are reached within 30 to 120 minutes (median 60 minutes) of oral dosing in the fasted state. The mean absolute oral bioavailability is 41% (range 25-63%). After oral three times a day dosing of Sildenafil, AUC and C_{max} increase in proportion with dose over the dose range of 20-40 mg. After oral doses of 80 mg three times a day a more than dose proportional increase in Sildenafil plasma levels has been observed. In pulmonary arterial hypertension patients, the oral bioavailability of Sildenafil after 80 mg three times a day was on average 43% (90% CI: 27-60%) higher compared to the lower doses.
When Sildenafil is taken with food, the rate of absorption is reduced with a mean delay in T_{max} of 60 minutes and a mean reduction in C_{max} of 29% however, the extent of absorption was not significantly affected (AUC decreased by 11%).

Distribution

The mean steady state volume of distribution (V_{SS}) for Sildenafil is 105 L, indicating distribution into the tissues. After oral doses of 20 mg three times a day, the mean maximum total plasma concentration of Sildenafil at steady state is approximately 113 ng/mL. Sildenafil and its major circulating N-desmethyl metabolite are approximately 96% bound to plasma proteins. Protein binding is independent of total drug concentrations.

Biotransformation

Sildenafil is cleared predominantly by the CYP3A4 (major route) and CYP2C9 (minor route) hepatic microsomal isoenzymes. The major circulating metabolite results from N-demethylation of Sildenafil. This metabolite has a phosphodiesterase selectivity profile similar to Sildenafil and an *in vitro* potency for PDE5 approximately 50% that of the parent drug. The N-desmethyl metabolite is further metabolized, with a terminal half-life of approximately 4 h. In patients with pulmonary arterial hypertension, plasma concentrations of N-desmethyl metabolite are approximately 72% those of Sildenafil after 20 mg three times a day dosing (translating into a 36% contribution to Sildenafil's pharmacological effects). The subsequent effect on efficacy is unknown.

Elimination

The total body clearance of Sildenafil is 41 L/h with a resultant terminal phase half-life of 3-5 h. After either oral or intravenous administration, Sildenafil is excreted as metabolites predominantly in the feces (approximately 80% of administered oral dose) and to a lesser extent in the urine (approximately 13% of administered oral dose).

Pharmacokinetics in special patient groups

Elderly

Healthy elderly volunteers (65 years or over) had a reduced clearance of Sildenafil, resulting in approximately 90% higher plasma concentrations of Sildenafil and the active N-desmethyl metabolite compared to those seen in healthy younger volunteers (18-45 years). Due to age-differences in plasma protein binding, the corresponding increase in free Sildenafil plasma concentration was approximately 40%.

Renal insufficiency

In volunteers with mild to moderate renal impairment (creatinine clearance = 30-80 mL/min), the pharmacokinetics of Sildenafil were not altered after receiving a 50 mg single oral dose. In volunteers with severe renal impairment (creatinine clearance < 30 mL/min), Sildenafil clearance was reduced, resulting in mean increases in AUC and C_{max} of 100% and 88% respectively compared to age-matched volunteers with no renal impairment. In addition, N-desmethyl metabolite AUC and C_{max} values were significantly increased by 200% and 79% respectively in subjects with severe renal impairment compared to subjects with normal renal function.

Hepatic insufficiency

In volunteers with mild to moderate hepatic cirrhosis (Child-Pugh class A and B) Sildenafil clearance was reduced, resulting in increases in AUC (85%) and C_{max} (47%) compared to age-matched volunteers with no hepatic impairment. In addition, N-desmethyl metabolite AUC and C_{max} values were significantly increased by 154% and 87%, respectively in cirrhotic subjects compared to subjects with normal hepatic function. The pharmacokinetics of Sildenafil in patients with severely impaired hepatic function have not been studied.

Population pharmacokinetics

In patients with pulmonary arterial hypertension, the average steady state concentrations were 20-50% higher over the investigated dose range of 20–80 mg three times a day compared to healthy volunteers. There was a doubling of the C_{min} compared to healthy volunteers. Both findings suggest a lower clearance and/or a higher oral bioavailability of Sildenafil in patients with pulmonary arterial hypertension compared to healthy volunteers.

Pediatric population

Pharmacokinetic profile of Sildenafil in the pediatric patients, body weight was shown to be a good predictor of drug exposure in children. Sildenafil plasma concentration half-life values were estimated to range from 4.2 to 4.4 hours for a range of 10 to 70 Kg of body weight and did not show any differences that would appear as clinically relevant. C_{max} after a single 20 mg Sildenafil dose administered PO was estimated at 49, 104 and 165 ng/mL for 70, 20 and 10 Kg patients, respectively. C_{max} after a single 10 mg Sildenafil dose administered PO was estimated at 24, 53 and 85 ng/mL for 70, 20 and 10 Kg patients, respectively. T_{max} was estimated at approximately 1 hour and was almost independent from body weight.

INDICATION:

Treatment of adult patients with pulmonary arterial hypertension classified as WHO functional class II and III, to improve exercise capacity. Efficacy has been shown in primary pulmonary hypertension and pulmonary hypertension associated with connective tissue disease.

DOSAGE AND ADMINISTRATION:

Oral

Tablets should be taken approximately 6 to 8 hours apart with or without food.

Treatment should only be initiated and monitored by a physician experienced in the treatment of pulmonary arterial hypertension. In case of clinical deterioration in spite of Sildenafil treatment, alternative therapies should be considered.

Adults (> 18 years)

The recommended dose is 20 mg three times a day (TID). Physicians should advise patients who forget to take Sildenafil to take a dose as soon as possible and then continue with normal dose. Patients should not take a double dose to compensate for the missed dose.

Patients using other medicinal products

In general, any dose adjustment should be administered only after a careful benefit-risk assessment. A downward dose adjustment to 20 mg twice daily should be considered when Sildenafil is co-administered to patients already receiving CYP3A4 inhibitors like erythromycin or Saquinavir. A downward dose adjustment to 20 mg once daily is recommended in case of co-administration with more potent CYP3A4 inhibitors Clarithromycin, Telithromycin and Nefazodone. For the use of Sildenafil with the most potent CYP3A4 inhibitors, see Contraindication. Dose adjustments of Sildenafil may be required when co-administered with CYP3A4 inducers (see Drug interaction).

Special populations

Elderly (≥65 years)

Dose adjustments are not required in elderly patients. Clinical efficacy as measured by 6-minute walk distance could be less in elderly patients.

Renal impairment

Initial dose adjustments are not required in patients with renal impairment, including severe renal impairment (creatinine clearance < 30 mL/min). A downward dose adjustment to 20 mg twice daily should be considered after a careful benefit-risk assessment only if therapy is not well-tolerated.

Hepatic impairment

Initial dose adjustments are not required in patients with hepatic impairment (Child-Pugh class A and B). A downward dose adjustment to 20 mg twice daily should be considered after a careful benefit-risk assessment only if therapy is not well-tolerated.
Sildenafil is contraindicated in patients with severe hepatic impairment (Child-Pugh class C) (see Contraindication).

Use in children

Sildenafil is not recommended for use in children below 18 years due to insufficient data on safety and efficacy.

Discontinuation of treatment

Limited data suggests that the abrupt discontinuation of Sildenafil is not associated with rebound worsening of pulmonary arterial hypertension. However, to avoid the possible occurrence of sudden clinical deterioration during withdrawal, a gradual dose reduction should be considered. Intensified monitoring is recommended during the discontinuation period.

PRECAUTION:

The efficacy of Sildenafil has not been established in patients with severe pulmonary arterial hypertension (functional class IV). If the clinical situation deteriorates, therapies that are recommended at the severe stage of the disease (e.g. Epoprostenol) should be considered (see Dosage and administration).
The benefit-risk balance of Sildenafil has not been established in patients assessed to be at WHO functional class I pulmonary arterial hypertension.

Retinitis pigmentosa

The safety of Sildenafil has not been studied in patients with known hereditary degenerative retinal disorders, such as *Retinitis pigmentosa* (minorities of these patients have genetic disorders of retinal phosphodiesterases) and therefore, its use is not recommended.

Vasodilatory action

When prescribing Sildenafil, physicians should carefully consider whether patients with certain underlying conditions could be adversely affected by Sildenafil's mild to moderate vasodilatory effects, for example patients with hypotension, patients with fluid depletion, severe left ventricular outflow obstruction or autonomic dysfunction (see Precaution).

Cardiovascular risk factors

Serious cardiovascular events, including myocardial infarction, unstable angina, sudden cardiac death, ventricular arrhythmia, cerebrovascular hemorrhage, transient ischemic attack, hypertension and hypotension have been reported in temporal association with the use of Sildenafil. Most, but not all, of these patients had pre-existing cardiovascular risk factors. Many events were reported to occur during or shortly after sexual intercourse and a few were reported to occur shortly after the use of Sildenafil without sexual activity. It is not possible to determine whether these events are related directly to these factors or to other factors.

Priapism

Sildenafil should be used with caution in patients with anatomical deformation of the penis (such as angulation, cavernosal fibrosis or Peyronie's disease), or in patients who have conditions which may predispose them to priapism (such as sickle cell anemia, multiple myeloma or leukemia).
Prolonged erections and priapism have been reported with Sildenafil in post-marketing experience. In the event of an erection that persists longer than 4 hours, the patient should seek immediate medical assistance. If priapism is not treated immediately, penile tissue damage and permanent loss of potency could result (see Side effect).

Vaso-occlusive crises in patients with sickle cell anemia

Sildenafil should not be used in patients with pulmonary hypertension secondary to sickle cell anemia.

Visual events

In the event of any sudden visual defect, the treatment should be stopped immediately and alternative treatment should be considered. (see Contraindication)

Alpha-blockers

Caution is advised when Sildenafil is administered to patients taking an alpha-blocker as the co-administration may lead to symptomatic hypotension in susceptible individuals (see Drug interaction). In order to minimize the potential for developing postural hypotension, patients should be hemodynamically stable on alpha-blocker therapy prior to initiating Sildenafil treatment. Physicians should advise patients what to do in the event of postural hypotensive symptoms.

Bleeding disorders

There is no safety information on the administration of Sildenafil to patients with bleeding disorders or active peptic ulceration. Therefore, Sildenafil should be administered to these patients only after careful benefit-risk assessment.

Vitamin K antagonists

In pulmonary arterial hypertension patients, there may be a potential for increased risk of bleeding when Sildenafil is initiated in patients already using a Vitamin K antagonist, particularly in patients with pulmonary arterial hypertension secondary to connective tissue disease.

Veno-occlusive disease

No data are available with Sildenafil in patients with pulmonary hypertension associated with pulmonary veno-occlusive disease. However, cases of life-threatening pulmonary edema have been reported with vasodilators (mainly Prostacyclin) when used in those patients. Consequently, should signs of pulmonary edema occur when Sildenafil is administered in patients with pulmonary hypertension, the possibility of associated veno-occlusive disease should be considered.

Galactose intolerance

Lactose monohydrate is present in the tablet. Patients with rare hereditary problems of galactose intolerance, the Lapp-lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Source of LUDIPRESS

Lactose monohydrate in Ludipress is a by-product.

Use of Sildenafil with Bosentan

The efficacy of Sildenafil in patients already on Bosentan therapy has not been conclusively demonstrated.

Concomitant use with other PDE5 inhibitors

The safety and efficacy of Sildenafil when co-administered with other PDE5 inhibitor products has not been studied in PAH patients and such concomitant use is not recommended (see Drug interaction).

PREGNANCY AND LACTATION:

Women of childbearing potential and contraception in males and females

Due to lack of data on effects of Sildenafil in pregnant women, Sildenafil is not recommended for women of childbearing potential unless also using appropriate contraceptive measures.

Pregnancy

There are no data from the use of Sildenafil in pregnant women.
Due to lack of data, Sildenafil should not be used in pregnant women unless strictly necessary.

Breastfeeding

Data from one lactating woman indicate that Sildenafil and its active metabolite N-desmethylsildenafil are excreted into breast milk at very low levels. No clinical data are available regarding adverse events in breastfed infants, but amounts ingested would not be expected to cause any adverse effects. Prescribers should carefully assess the mother's clinical need for Sildenafil and any potential adverse effects on the breastfed child.

Fertility

Non-clinical data revealed no special hazard for humans based on conventional studies of fertility.

SIDE EFFECT:

Tabulated list of adverse reactions

MedDRA System Organ Class (V.14.0)	Adverse reaction
Infections and infestations Common	 Cellulitis, influenza, bronchitis, sinusitis, rhinitis, gastroenteritis
Blood and lymphatic system disorders Common	 Anemia
Metabolism and nutrition disorders Common	 Fluid retention
Psychiatric disorders Common	 Insomnia, anxiety
Nervous system disorders Very common Common	 Headache Migraine, tremor, paresthesia, burning sensation, hypoesthesia
Eye disorders Common Uncommon Not known	 Retinal hemorrhage, visual impairment, vision blurred, photophobia, chromatopsia, cyanopsia, eye irritation, ocular hyperemia Visual acuity reduced, diplopia, abnormal sensation in eye *Non-arteritic anterior ischemic optic neuropathy (NAION)*, Retinal vascular occlusion*, Visual field defect**
Ear and labyrinth disorders Common Not known	 Vertigo *Sudden hearing loss*
Vascular disorders Very common Not Known	 Flushing *Hypotension*

Respiratory, thoracic and mediastinal disorders Common	 Epistaxis, cough, nasal congestion
Gastrointestinal disorders Very common Common	 Diarrhea, dyspepsia Gastritis, gastroesophageal reflux disease, hemorrhoids, abdominal distension, dry mouth
Skin and subcutaneous tissue disorders Common Not known	 Alopecia, erythema, night sweats *Rash*
Musculoskeletal and connective tissue disorders Very common Common	 Pain in extremity Myalgia, back pain
Renal and urinary disorders Uncommon	 Hematuria
Reproductive system and breast disorders Uncommon Not known	 Penile hemorrhage, hematospermia, gynecomastia *Priapism, erection increased*
General disorders and administration site conditions Common	 Pyrexia

*These adverse events/reactions have been reported in patients taking Sildenafil for the treatment of male erectile dysfunction (MED).
Reports from post-marketing experience are included in italics.

EFFECT ON ABILITY TO DRIVE AND USE MACHINES:

Sildenafil has moderate influence on the ability to drive and use machines.
As dizziness and altered vision were reported with Sildenafil, patients should be aware of how they might be affected by Sildenafil, before driving or using machines.

CONTRAINDICATION:

- Hypersensitivity to the active substance or to any of the excipients
- Co-administration with Nitric oxide donors (such as Amyl nitrite) or Nitrates in any form due to the hypotensive effects of Nitrates
- The co-administration of PDE5 inhibitors, including Sildenafil, with Guanylate cyclase stimulators, such as Riociguat, is contraindicated as it may potentially lead to symptomatic hypotension.
- Combination with the most potent of the CYP3A4 inhibitors (e.g. Ketoconazole, Itraconazole, Ritonavir) (see Drug interaction)
- Patients who have loss of vision in one eye because of non-arteritic anterior ischemic optic neuropathy (NAION), regardless of whether this episode was in connection or not with previous PDE5 inhibitor exposure
- The safety of Sildenafil has not been studied in the following sub-groups of patients and its use is therefore contraindicated:
  - Severe hepatic impairment,
  - Recent history of stroke or myocardial infarction,
  - Severe hypotension (blood pressure < 90/50 mmHg) at initiation

DRUG INTERACTION:

Effects of other medicinal products on Sildenafil

Sildenafil metabolism is principally mediated by the cytochrome P450 (CYP) isoforms 3A4 (major route) and 2C9 (minor route). Therefore, inhibitors of these isoenzymes may reduce Sildenafil clearance and inducers of these isoenzymes may increase Sildenafil clearance.
The efficacy and safety of Sildenafil co-administered with other treatments for pulmonary arterial hypertension (e.g. Ambrisentan, Iloprost) has not been studied. Therefore, caution is recommended in case of co-administration.
The efficacy and safety of Sildenafil when co-administered with other PDE5 inhibitors has not been studied in pulmonary arterial hypertension patients.
Reduction in Sildenafil clearance and/or an increase of oral bioavailability when co-administered with CYP3A4 substrates and the combination of CYP3A4 substrates and beta-blockers.
CYP3A4 inducers seemed to have a substantial impact on the pharmacokinetics of Sildenafil in pulmonary arterial hypertension patients, which was confirmed with a CYP3A4 inducer Bosentan.
Efficacy of Sildenafil should be closely monitored in patients using concomitant potent CYP3A4 inducers, such as Carbamazepine, Phenytoin, Phenobarbital, St. John's wort and Rifampicin.
Co-administration of the HIV protease inhibitor Ritonavir, which is highly potent P450 inhibitor, at steady state with Sildenafil resulted in a 4-fold increase in Sildenafil C_{max} and an 11-fold increase in Sildenafil plasma AUC. Based on these pharmacokinetic results co-administration of Sildenafil with Ritonavir is contraindicated in pulmonary arterial hypertension patients.
Saquinavir, CYP3A4 inhibitor, at steady state with Sildenafil resulted in 140% increase in Sildenafil C_{max} and a 210% increase in Sildenafil AUC. Sildenafil had no effect on Saquinavir pharmacokinetics.
Erythromycin, moderate CYP3A4 inhibitor at steady state, there was a 182% increase in Sildenafil systemic exposure (AUC).
No dose adjustment is required for Azithromycin and Cimetidine.
The most potent of the CYP3A4 inhibitors such as Ketoconazole and Itraconazole would be expected to have effects similar to Ritonavir. CYP3A4 inhibitors like Clarithromycin, Telithromycin and Nefazodone are expected to have an effect between that of Ritonavir and CYP3A4 inhibitors like Saquinavir or Erythromycin. Therefore, dose adjustments are recommended when using CYP3A4 inhibitors.
Nicorandil is a hybrid of potassium channel activator and nitrate. Due to the nitrate component it has the potential to have serious interaction with Sildenafil.

Effects of Sildenafil on other medicinal products

Sildenafil when co-administered with Amlodipine in hypertensive patients had an additional reduction on supine systolic blood pressure.
When Sildenafil and Doxazosin were administered simultaneously to patients stabilized on Doxazosin therapy, there were infrequent reports of patients who experienced symptomatic postural hypotension. These reports included dizziness and lightheadedness, but not syncope. Concomitant administration of Sildenafil to patients taking alpha-blocker therapy may lead to symptomatic hypotension in susceptible individuals.
Consistent with its known effects on the nitric oxide/cGMP pathway, Sildenafil was shown to potentiate the hypotensive effects of nitrates, and its co-administration with Nitric oxide donors or nitrates in any form is therefore contraindicated.

Pediatric population

Interaction studies have only been performed in adults.

OVERDOSE AND TREATMENT:

Doses up to 800 mg, adverse reactions were similar to those seen at lower doses, but the incidence rates and severities were increased. At single doses of 200 mg the incidence of adverse reactions (headache, flushing, dizziness, dyspepsia, nasal congestion, and altered vision) was increased.
In cases of overdose, standard supportive measures should be adopted as required. Renal dialysis is not expected to accelerate clearance as Sildenafil is highly bound to plasma proteins and not eliminated in the urine.

STORAGE:

Store at temperature of not more than 30ºC. Store in the original package in order to protect from moisture.

DOSAGE FORM AND PACKAGING AVAILABLE:

20 mg Tablet, Blister 10x10's

DATE OF REVISION:

June 29, 2023

Manufactured by:
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